

NovaDesk AI

Project Documentation

AI-Powered Customer Support Assistant
with Live Response Guidance

7+	<100ms	3	7	Real-time
AI Agents	Provisional	Modes	Languages	WS

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Built with FastAPI, LangGraph, ChromaDB RAG, React, and Tailwind CSS

NovaDesk AI -- Coach at every turn.

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1. Project Overview

NovaDesk AI is a multi-agent, real-time coaching platform that helps customer support agents perform better during live conversations. Every interaction is analyzed on the fly by a LangGraph-powered pipeline -- intent, sentiment, escalation risk, knowledge retrieval, and a suggested reply -- before the customer even finishes typing.

The system acts as an intelligent coach that sits alongside support agents, providing instant feedback, suggested responses, and escalation alerts during real-time customer interactions.

Key Highlights

- * Real-time WebSocket-based coaching during live conversations
- * Multi-agent AI pipeline orchestrated via LangGraph
- * RAG-powered knowledge base with ChromaDB vector store
- * Three interaction modes: Simulator, Manual, and Replay
- * Role-based access control (Admin, Trainer, Agent)
- * PDF report generation with coaching analytics

- * Works offline with rule-based fallback when no LLM key is configured

2. Problem Statement

Customer support agents face several challenges:

Pain Point	Detail
Lack of real-time guidance	Agents receive coaching only after calls end (post-call reviews) -- reactive and slow
High cognitive load	Juggling multiple conversations while maintaining quality is exhausting
Inconsistent response quality	Without immediate feedback, agents may provide suboptimal replies
Delayed escalation	Critical customer issues may not be flagged until it's too late
Limited practice opportunities	New agents lack safe environments to practice before handling real customers

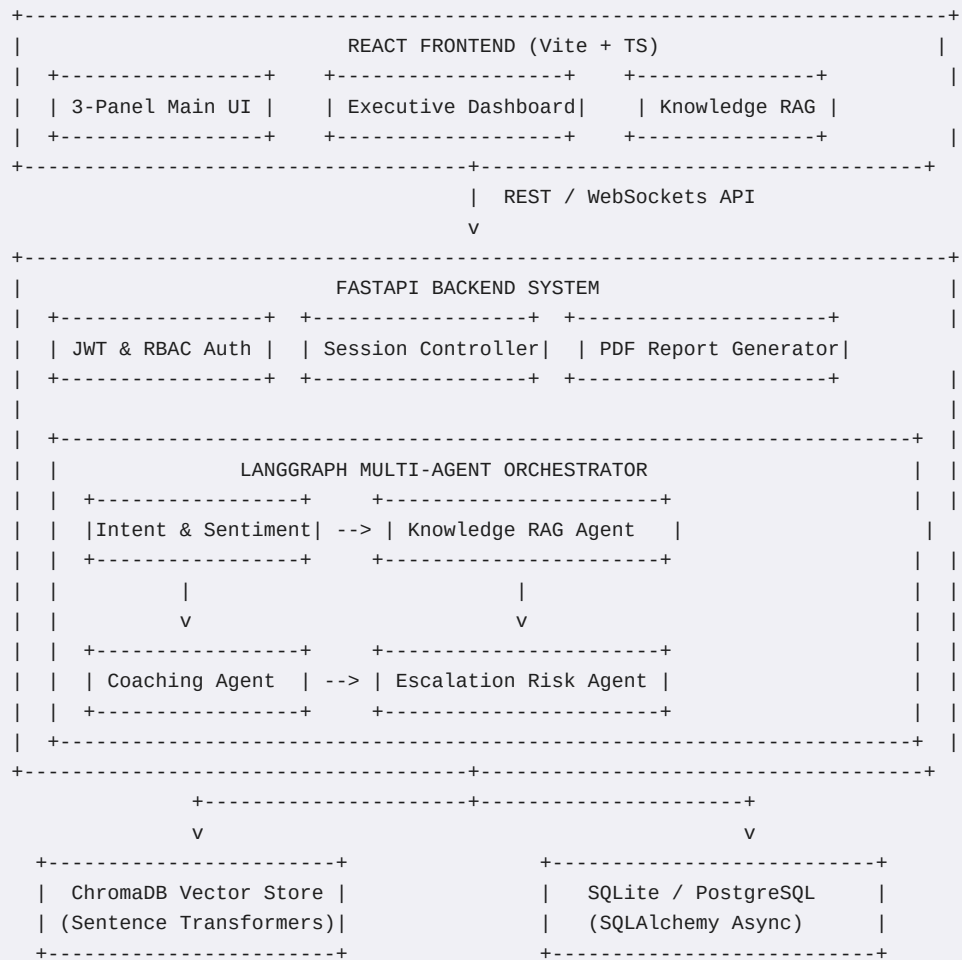
3. Proposed Solution

NovaDesk AI addresses these challenges through:

- * Live coaching engine -- A multi-agent AI pipeline that analyzes every conversation turn in real-time, providing instant scores (tone, grammar, empathy) and suggested replies.
- * Customer simulator -- AI-powered customer personas that generate realistic conversations for agent training.
- * RAG knowledge base -- Upload and ingest support documents (PDF, DOCX, TXT) into a ChromaDB vector store for contextual knowledge retrieval during conversations.
- * Escalation monitoring -- Automatic detection of escalation risk levels (Low, Medium, High, Critical) with recommended supervisor actions.
- * Post-interaction analytics -- Detailed coaching reports with sentiment journeys, score trends, and downloadable PDF summaries.

4. System Architecture

High-Level Architecture



Data Flow

1. User sends message -> Frontend via REST API or WebSocket
2. Orchestrator receives request -> Routes through LangGraph pipeline
3. Knowledge RAG Agent -> Queries ChromaDB for relevant support articles
4. Combined Analysis Agent -> Single LLM call returns intent, sentiment, coaching scores, escalation risk, and suggested reply
5. Response persisted -> Stored in SQLite via SQLAlchemy
6. Coaching feedback sent -> Returned to frontend in real-time

5. Features

5.1 Live Coaching (WebSocket)

- * Real-time conversation streaming with live agent guidance over WebSocket
- * Two-phase analysis: instant rule-based provisional result (< 100ms), followed by LLM refinement in

background

- * Per-turn scores for tone, grammar, empathy, confidence
- * Ready-to-send suggested reply with reasoning and improvement tips

5.2 Multi-Agent Pipeline

Agent	Responsibility
Customer Simulator Agent	Generates persona-consistent customer turns (6 personas)
Intent & Sentiment Agent	Classifies intent, emotion, sentiment, urgency, frustration (0.0-1.0)
Combined Analysis Agent	Single-pass intent + sentiment + coaching + escalation analysis
Knowledge Recommendation Agent	RAG retrieval over ChromaDB with cosine similarity search
Coaching & Response Agent	Scores drafts and generates the ideal reply
Escalation Risk Monitor	Flags Low / Medium / High / Critical risk with actions
Post-Interaction Summary Agent	Generates structured coaching summaries and PDF reports

5.3 Interaction Modes

Mode	Description
Simulator Mode	Practice against 6 realistic AI customer personas with emotional progression
Manual Mode	Paste live customer messages and receive immediate coaching
Replay Mode	Analyze past transcripts turn by turn with full coaching insights

5.4 Knowledge Base Management

- * Upload PDF, DOCX, TXT, MD documents
- * Auto-chunked (800 chars, 150 overlap) and embedded into ChromaDB
- * Semantic search with cosine similarity
- * Document versioning and deletion

5.5 Analytics & Reporting

- * Executive metrics dashboard with trend charts
- * Sentiment journey visualization per session
- * Score trends over time
- * Downloadable PDF coaching reports

5.6 Accessibility

- * Multi-language translation (Hindi, Hinglish, Kannada, Telugu, Tamil, Marathi, Bengali)
- * Read-aloud functionality for suggested replies
- * Dark/Light theme support

6. Technology Stack

Backend

Layer	Technology	Purpose
API Framework	FastAPI + Uvicorn	Async REST API and WebSocket server
Data Validation	Pydantic v2	Request/response schemas
Orchestration	LangGraph	Multi-agent workflow graph
LLM Integration	OpenAI / Google Gemini / Groq	AI model inference
Vector Store	ChromaDB	RAG document storage and retrieval
Database	SQLAlchemy 2.0 +aiosqlite	Async SQLite ORM
Authentication	JWT (python-jose) + passlib/bcrypt	Secure auth with RBAC
Reports	ReportLab	PDF generation
Real-time	WebSockets (FastAPI)	Live coaching streaming

Frontend

Layer	Technology	Purpose
Framework	React 19 + TypeScript	UI components
Build Tool	Vite 8	Fast dev server and bundler
Styling	Tailwind CSS v4	Glass UI, animations
State Management	Zustand	Global client state
Data Fetching	TanStack Query	Server state caching
Charts	Recharts	Analytics visualizations
Routing	React Router v7	SPA navigation
Motion	Framer Motion	UI animations

DevOps

Tool	Purpose
Docker + Docker Compose	Containerized deployment
Nginx	Reverse proxy for production
Render	Cloud hosting platform

7. Database Design

Table	Description
users	Stores user accounts with roles (admin, trainer, agent)
sessions	Coaching sessions with mode, persona, scenario, and status
messages	Individual conversation messages within sessions
coaching_analysis	Per-turn AI analysis results (intent, sentiment, scores, suggested reply)
coaching_feedback	User feedback on coaching suggestions (helpful/not helpful)
documents	Uploaded knowledge base documents metadata
reports	Post-interaction coaching summaries and analytics
system_settings	Application configuration key-value pairs

Escalation Risk Levels

Level	Condition	Action
Low	Normal conversation, positive/neutral sentiment	Standard support dialogue
Medium	Mild frustration or confusion	Provide empathetic reassurance
High	detected High frustration + urgent dissatisfaction	Specialist transfer or manager notification
Critical	Supervisor requested, legal mentions	Immediate manager escalation

8. Multi-Agent Pipeline

LangGraph Workflow

```
# Simplified orchestration flow
knowledge_rag -> intent_sentiment -> END

# The pipeline consists of two main nodes:
# 1. Knowledge RAG Node -- Fast, no LLM call. Retrieves relevant support articles.
# 2. Combined Analysis Node -- Single LLM call returning all analysis.
```

Combined Analysis Agent Output

```
{
  "intent": "Technical Issue",
  "sentiment": "Negative",
  "emotion": "Frustrated / Angry",
  "urgency": "High",
  "frustration": 0.75,
  "tone_score": 82.5,
  "grammar_score": 91.0,
  "empathy_score": 78.0,
  "suggested_reply": "I understand your frustration...",
  "escalation_risk": "High",
  "recommended_action": "Transfer to Tier 2"
}
```

Fallback Mechanism

When no LLM API key is configured, the system falls back to deterministic rule-based logic:

- * Keyword matching for intent classification
- * Sentiment analysis via negative/urgent/stress keyword counting
- * Pre-defined suggested replies per intent category
- * Threshold-based escalation scoring

9. API Documentation

Authentication (/api/v1/auth)

Method	Endpoint	Description
POST	/auth/register	Register new user (admin, trainer, agent)
POST	/auth/login	Authenticate and receive JWT token
POST	/auth/forgot-password	Send password recovery instructions
GET	/auth/me	Get current user profile and role

Session Management (/api/v1/session)

Method	Endpoint	Description
POST	/session/create	Create coaching session
GET	/session/list	List user's coaching sessions
GET	/session/{id}	Get session messages and history
DELETE	/session/{id}	Delete coaching session

Chat & Live Coaching (/api/v1/chat)

Method	Endpoint	Description
POST	/chat/message	Process chat turn through multi-agent pipeline
POST	/chat/simulator-next	Generate next simulator persona message
POST	/chat/manual	Manual mode coaching evaluation
POST	/chat/replay	Replay pre-loaded transcript
WS	/chat/ws/{session_id}	Real-time WebSocket coaching stream

RAG Knowledge Base (/api/v1/rag)

Method	Endpoint	Description
POST	/rag/upload	Upload document (PDF, DOCX, TXT, MD)
POST	/rag/search	Semantic search over knowledge base
GET	/rag/documents	List ingested documents
DELETE	/rag/documents/{id}	Remove document from vector index

Reports & Analytics

Method	Endpoint	Description
POST	/report/generate/{session_id}	Generate post-interaction summary report
GET	/report/{session_id}/pdf	Download PDF report file
GET	/analytics/summary	Executive metrics and trend data

10. Frontend Architecture

Page Structure

Page	Route	Description
Landing Page	/	Public landing with multi-agent demo animation
Main Console	/console	3-panel coaching interface
Dashboard	/dashboard	Executive overview with key metrics
Sessions	/sessions	Coaching session history and management
Knowledge Base	/knowledge	Document upload and management
Analytics	/analytics	Charts and performance trends
Reports	/reports	Post-interaction coaching reports

Component Architecture

```
App.tsx
+-- QueryClientProvider (TanStack Query)
+-- BrowserRouter
  +-- LandingPage (public)
  +-- RegisterPage (public)
  +-- DashboardLayout (protected)
    +-- Sidebar
    +-- Header
    +-- [Page Content via Outlet]
      +-- MainConsolePage
        | +-- Conversation Panel
        | +-- Coaching Panel
        | +-- Knowledge Panel
      +-- DashboardPage
      +-- SessionsPage
      +-- KnowledgeBasePage
      +-- AnalyticsPage
      +-- ReportsPage
```

Real-Time Communication

The useCoachSocket custom hook manages WebSocket connections: auto-connects on session start, sends customer/agent messages, receives instant coaching analysis, handles background LLM refinement updates, and provides graceful reconnection on disconnect.

11. Installation & Setup

Prerequisites

- * Python 3.11+
- * Node.js 18+ and npm
- * LLM API key (optional) -- OpenAI, Google Gemini, or Groq

Backend Setup

```
cd customer-support-coach
python -m venv .venv
.venv\Scripts\activate          # Windows
source .venv/bin/activate       # macOS / Linux
pip install -r requirements.txt
copy .env.example .env          # Add your API keys
uvicorn backend.main:app --reload --port 8000
```

Frontend Setup

```
cd frontend
npm install
npm run dev                      # http://localhost:5173
```

Docker (One Command)

```
docker-compose -f docker/docker-compose.yml up --build
# Frontend: http://localhost:80
# Backend: http://localhost:8000/docs
```

Demo Accounts

Role	Email	Password
Agent	agent@coach.ai	Agent@123456
Trainer	trainer@coach.ai	Trainer@123456
Admin	admin@coach.ai	Admin@123456

12. Configuration

Backend Environment Variables

Variable	Default	Description
SECRET_KEY	change-in-prod	JWT signing secret
DATABASE_URL	sqlite+aiosqlite:///./sql_app.db	Database connection string
OPENAI_API_KEY	(empty)	OpenAI API key
GEMINI_API_KEY	(empty)	Google Gemini API key
GROQ_API_KEY	(empty)	Groq API key
LLM_MODEL	gpt-4o	Default LLM model
EMBEDDING_MODEL	all-MiniLM-L6-v2	RAG embedding model
CHUNK_SIZE	800	Document chunk size
CHUNK_OVERLAP	150	Chunk overlap size

Frontend Environment Variables

Variable	Default	Description
VITE_API_URL	http://localhost:8000/api/v1	Backend REST base URL
VITE_WS_URL	ws://localhost:8000	Backend WebSocket base URL

13. Project Structure

```
customer-support-coach/
+-- backend/
|   +-- agents/                # Multi-agent implementation
|   |   +-- orchestrator.py    # LangGraph workflow orchestrator
|   |   +-- combined_agent.py  # Single-pass analysis
|   |   +-- coaching_agent.py  # Response scoring & suggestions
|   |   +-- knowledge_agent.py # RAG retrieval agent
|   |   +-- simulator_agent.py # Customer persona generator
|   +-- api/                   # FastAPI route handlers
|   +-- authentication/        # JWT + RBAC
|   +-- core/                   # LLM service, config
|   +-- database/               # SQLAlchemy models
|   +-- rag/                    # ChromaDB vector store
|   +-- reports/                # PDF generation
+-- frontend/
|   +-- src/pages/              # React page components
|   +-- src/components/         # Shared UI components
|   +-- src/store/              # Zustand state
|   +-- src/hooks/              # useCoachSocket
+-- docker/                     # Docker configs
+-- docs/                       # Project documentation
+-- tests/                      # Backend test suite
+-- data/                       # Knowledge base, transcripts
```

14. Testing

```
cd customer-support-coach
pytest
pytest tests/test_backend.py
pytest tests/test_agents.py
pytest tests/test_rag.py
pytest tests/test_websocket_chat.py
```

Test File	Coverage Area
test_backend.py	API endpoint tests (auth, session, chat, rag, report)
test_agents.py	Agent logic tests (combined, coaching, simulator, escalation)
test_rag.py	RAG pipeline tests (chunking, ingestion, vector search)
test_backend_rag.py	Integration tests for RAG + API
test_websocket_chat.py	WebSocket real-time coaching tests

15. Deployment

Docker Deployment

```
docker-compose -f docker/docker-compose.yml up --build
# Backend (FastAPI): http://localhost:8000
# Frontend (React): http://localhost:80
```

Render Cloud Deployment

The project includes render.yaml for one-click deployment on Render.

Production Checklist

- * Change SECRET_KEY to a secure random value
- * Set proper CORS origins (replace * with domain)
- * Use PostgreSQL instead of SQLite
- * Configure LLM API keys
- * Set up SSL/TLS certificates
- * Configure proper logging and monitoring

16. Future Scope

1. Fine-tuned Models -- Train domain-specific models on historical support data
2. Multi-language Support -- Real-time coaching in multiple languages
3. Browser Extension -- Chrome extension for live coaching on any support platform
4. Integration APIs -- Connect with Zendesk, Freshdesk, Intercom, Salesforce
5. Voice Coaching -- Real-time audio analysis and voice tone coaching
6. Team Analytics -- Manager dashboards with team performance comparisons
7. Gamification -- Achievement badges, leaderboards, and training milestones
8. Advanced RAG -- Hybrid search (keyword + semantic), query expansion, reranking
9. A/B Testing -- Compare coaching strategies across agent groups
10. Mobile App -- Native iOS/Android app for coaching on the go